Hospital Management System

-- By Sahil Kadam

Show All Tables :-

- Select * from Patients;
- Select * from Doctors;
- Select * from Appointments;

Patients :-

	patient_id	name	age	gender	city
1	1	Rahul Nair	25	Male	Kochi
2	2	Anita Sharma	30	Female	Delhi
3	3	Vivek Kumar	45	Male	Chennai
4	4	Sneha Menon	28	Female	Trivandrum
5	5	Arjun Singh	50	Male	Mumbai
6	6	Divya Reddy	35	Female	Hyderabad
7	7	Kiran Patel	40	Male	Ahmedabad
8	8	Meera Das	22	Female	Kolkata
9	9	Suresh Babu	60	Male	Madurai
10	10	Neha Kapoor	27	Female	Chandigarh
11	11	Manoj Pillai	33	Male	Thrissur
12	12	Lakshmi Iyer	48	Female	Coimbatore
13	13	Amit Verma	38	Male	Bangalore
14	14	Priya Joshi	29	Female	Pune
15	15	Rohit Mehta	55	Male	Jaipur

Doctors :-

	doctor_id	name	specialization	phone
1	101	Dr. Anil Kumar	Cardiologist	9000011111
2	102	Dr. Priya Menon	Dematologist	9000011112
3	103	Dr. Rajesh Sharma	Orthopedic	9000011113
4	104	Dr. Meena Reddy	Pediatrician	9000011114
5	105	Dr. Sunil Das	Neurologist	9000011115

Appointments :-

	appointment_id	patient_id	doctor_id	appointment_date	diagnosis
1	1001	1	101	2025-02-01	Chest Pain
2	1002	2	102	2025-02-02	Skin Allergy
3	1003	3	103	2025-02-03	Fracture
4	1004	4	104	2025-02-04	Fever
5	1005	5	101	2025-02-05	Heart Checkup
6	1006	6	105	2025-02-06	Migraine
7	1007	7	103	2025-02-07	Back Pain
8	1008	8	104	2025-02-08	Cold & Cough

Query 1 :-

- Patients older than 40

Select * from Patients

Where age > 40;

	patient_id	name	age	gender	city
1	3	Vivek Kumar	45	Male	Chennai
2	5	Arjun Singh	50	Male	Mumbai
3	9	Suresh Babu	60	Male	Madurai
4	12	Lakshmi Iyer	48	Female	Coimbatore
5	15	Rohit Mehta	55	Male	Jaipur

Query 2 :-

- Female Patients above age 30

Select * from Patients

Where gender = ‘Female’ AND age > 30;

	patient_id	name	age	gender	city
1	6	Divya Reddy	35	Female	Hyderabad
2	12	Lakshmi Iyer	48	Female	Coimbatore

Query 3 :-

- Doctors who are Cardiologists OR Neurologist

Select * from Doctors

Where specialization = ‘Cardiologists’ OR specialization = 'Neurologist' ;

	doctor_id	name	specialization	phone
1	101	Dr. Anil Kumar	Cardiologist	9000011111
2	105	Dr. Sunil Das	Neurologist	9000011115

Query 4 :-

- Appointments after Feb 5

Select * from Appointments

Where appointment_date > ‘2025-02-05’;

	appointment_id	patient_id	doctor_id	appointment_date	diagnosis
1	1006	6	105	2025-02-06	Migraine
2	1007	7	103	2025-02-07	Back Pain
3	1008	8	104	2025-02-08	Cold & Cough

Query 5 :-

- Patients between age 30 and 50

Select * FROM Patients

Where age BETWEEN 30 AND 50;

	patient_id	name	age	gender	city
1	2	Anita Sharma	30	Female	Delhi
2	3	Vivek Kumar	45	Male	Chennai
3	5	Arjun Singh	50	Male	Mumbai
4	6	Divya Reddy	35	Female	Hyderabad
5	7	Kiran Patel	40	Male	Ahmedabad
6	11	Manoj Pillai	33	Male	Thrissur
7	12	Lakshmi Iyer	48	Female	Coimbatore
8	13	Amit Verma	38	Male	Bangalore

Query 6 :-

- Patients name start with ‘A’

Select * from Patients

Where name LIKE ‘A%’;

	patient_id	name	age	gender	city
1	2	Anita Shama	30	Female	Delhi
2	5	Arjun Singh	50	Male	Mumbai
3	13	Amit Verma	38	Male	Bangalore

Query 7 :-

- Total patients

SELECT COUNT(*) AS Total_Patients FROM Patients;

	Total_Patients
1	15

- Average patient age

SELECT AVG(age) AS Avg_Age FROM Patients;

	Avg_Age
1	37

- Oldest patient

SELECT MAX(age) AS Max_Age FROM Patients;

	Max_Age
1	60

- Youngest patient

SELECT MIN(age) AS Min_Age FROM Patients;

	Min_Age
1	22

Query 8 :-

- Count patients by gender

```
SELECT gender, COUNT(*) as Total

From Patients

GROUP BY gender;
```

	gender	Total
1	Female	7
2	Male	8

Query 9 :-

- Number of appointments per doctor

```
SELECT doctor_id, COUNT(*) AS Total_Appointments
FROM Appointments
GROUP BY doctor_id;
```

	doctor_id	Total_Appointments
1	101	2
2	102	1
3	103	2
4	104	2
5	105	1

Query 10 :-

- Patients sorted by age

```
SELECT * FROM Patients

ORDER BY age ASC;
```

	patient_id	name	age	gender	city
1	8	Meera Das	22	Female	Kolkata
2	1	Rahul Nair	25	Male	Kochi
3	10	Neha Kapoor	27	Female	Chandigarh
4	4	Sneha Menon	28	Female	Trivandrum
5	14	Priya Joshi	29	Female	Pune
6	2	Anita Sharma	30	Female	Delhi
7	11	Manoj Pillai	33	Male	Thrissur
8	6	Divya Reddy	35	Female	Hyderabad
9	13	Amit Verma	38	Male	Bangalore
10	7	Kiran Patel	40	Male	Ahmedabad
11	3	Vivek Kumar	45	Male	Chennai
12	12	Lakshmi Iyer	48	Female	Coimbatore
13	5	Arjun Singh	50	Male	Mumbai
14	15	Rohit Mehta	55	Male	Jaipur
15	9	Suresh Babu	60	Male	Madurai

Query 11 :-

- Doctors with more than 1 appointment

```
SELECT doctor_id, COUNT(*) AS Total_Appointments
FROM Appointments
GROUP BY doctor_id
HAVING COUNT(*) > 1;
```

	doctor_id	Total_Appointments
1	101	2
2	103	2
3	104	2

Query 12 :-

- Show patient name, doctor name, diagnosis

```
SELECT
  P.name AS Patient_Name,
  D.name AS Doctor_Name,
  A.diagnosis
FROM Appointments A
JOIN Patients P ON A.patient_id = P.patient_id
JOIN Doctors D ON A.doctor_id = D.doctor_id;
```

	Patient_Name	Doctor_Name	diagnosis
1	Rahul Nair	Dr. Anil Kumar	Chest Pain
2	Anita Sharma	Dr. Priya Menon	Skin Allergy
3	Vivek Kumar	Dr. Rajesh Sharma	Fracture
4	Sneha Menon	Dr. Meena Reddy	Fever
5	Arjun Singh	Dr. Anil Kumar	Heart Checkup
6	Divya Reddy	Dr. Sunil Das	Migraine
7	Kiran Patel	Dr. Rajesh Sharma	Back Pain
8	Meera Das	Dr. Meena Reddy	Cold & Cough

Query 13 :-

- Show all doctors (even without appointments)

```
SELECT D.name, A.appointment_date

FROM Appointments A

RIGHT JOIN Doctors D

ON A.doctor_id = D.doctor_id;
```

	name	appointment_date
1	Dr. Anil Kumar	2025-02-01
2	Dr. Anil Kumar	2025-02-05
3	Dr. Priya Menon	2025-02-02
4	Dr. Rajesh Sharma	2025-02-03
5	Dr. Rajesh Sharma	2025-02-07
6	Dr. Meena Reddy	2025-02-04
7	Dr. Meena Reddy	2025-02-08
8	Dr. Sunil Das	2025-02-06

Query 14 :-

- Rank doctors by number of appointments

```
SELECT
  D.name, COUNT(A.appointment_id) AS Total_Appointments,
  RANK() OVER (ORDER BY COUNT(A.appointment_id) DESC) AS Rank_Position
FROM Doctors D
LEFT JOIN Appointments A
ON D.doctor_id = A.doctor_id
GROUP BY D.name;
```

	name	Total_Appointments	Rank_Position
1	Dr. Anil Kumar	2	1
2	Dr. Meena Reddy	2	1
3	Dr. Rajesh Sharma	2	1
4	Dr. Sunil Das	1	4
5	Dr. Priya Menon	1	4

Query 15 :-

- Dense Rank

```
SELECT name, age,
DENSE_RANK() OVER (ORDER BY age DESC) AS Dense_Rank_Position
FROM Patients;
```

	name	age	Dense_Rank_Position
1	Suresh Babu	60	1
2	Rohit Mehta	55	2
3	Arjun Singh	50	3
4	Lakshmi Iyer	48	4
5	Vivek Kumar	45	5
6	Kiran Patel	40	6
7	Amit Verma	38	7
8	Divya Reddy	35	8
9	Manoj Pillai	33	9
10	Anita Sharma	30	10
11	Priya Joshi	29	11
12	Sneha Menon	28	12
13	Neha Kapoor	27	13
14	Rahul Nair	25	14
15	Meera Das	22	15

Query 16 :-

- Patients older than average age

```
SELECT * FROM Patients

WHERE age > (SELECT AVG(age) FROM Patients);
```

	patient_id	name	age	gender	city
1	3	Vivek Kumar	45	Male	Chennai
2	5	Arjun Singh	50	Male	Mumbai
3	7	Kiran Patel	40	Male	Ahmedabad
4	9	Suresh Babu	60	Male	Madurai
5	12	Lakshmi Iyer	48	Female	Coimbatore
6	13	Amit Verma	38	Male	Bangalore
7	15	Rohit Mehta	55	Male	Jaipur

Query 17 :-

- Patients who have appointments

```
SELECT * FROM Patients
WHERE patient_id IN (
 SELECT patient_id FROM Appointments
);
```

	patient_id	name	age	gender	city
1	1	Rahul Nair	25	Male	Kochi
2	2	Anita Sharma	30	Female	Delhi
3	3	Vivek Kumar	45	Male	Chennai
4	4	Sneha Menon	28	Female	Trivandrum
5	5	Arjun Singh	50	Male	Mumbai
6	6	Divya Reddy	35	Female	Hyderabad
7	7	Kiran Patel	40	Male	Ahmedabad
8	8	Meera Das	22	Female	Kolkata

Query 18 :-

- Doctors who handled appointments in Feb 2025

```
SELECT * FROM Doctors
WHERE doctor_id IN (
    SELECT doctor_id
    FROM Appointments
    WHERE appointment_date BETWEEN '2025-02-01' AND '2025-02-28’
);
```

	doctor_id	name	specialization	phone
1	101	Dr. Anil Kumar	Cardiologist	9000011111
2	102	Dr. Priya Menon	Dermatologist	9000011112
3	103	Dr. Rajesh Sharma	Orthopedic	9000011113
4	104	Dr. Meena Reddy	Pediatrician	9000011114
5	105	Dr. Sunil Das	Neurologist	9000011115

Query 19 :-

- Patients whose name starts with ‘A’

```
SELECT * FROM Patients
WHERE REGEXP_LIKE (name, '^A');
```

	patient_id	name	age	gender	city
1	2	Anita Sharma	30	Female	Delhi
2	5	Arjun Singh	50	Male	Mumbai
3	13	Amit Verma	38	Male	Bangalore

Query 20 :-

- Patients whose name ends with ‘a’

```
SELECT * FROM Patients
WHERE REGEXP_LIKE (name, 'a$’);
```

	patient_id	name	age	gender	city
1	2	Anita Sharma	30	Female	Delhi
2	13	Amit Verma	38	Male	Bangalore
3	15	Rohit Mehta	55	Male	Jaipur

Query 21 :-

- Phone numbers containing only digits

```
SELECT * FROM Doctors

WHERE REGEXP_LIKE (phone, '^[0-9]+$');
```

	doctor_id	name	specialization	phone
1	101	Dr. Anil Kumar	Cardiologist	9000011111
2	102	Dr. Priya Menon	Dermatologist	9000011112
3	103	Dr. Rajesh Sharma	Orthopedic	9000011113
4	104	Dr. Meena Reddy	Pediatrician	9000011114
5	105	Dr. Sunil Das	Neurologist	9000011115

Query 22 :-

- Names containing 'ee’

```
SELECT * FROM Patients

WHERE name LIKE '%ee%';
```

	patient_id	name	age	gender	city
1	8	Meera Das	22	Female	Kolkata

Query 23 :-

- Update Information

```
UPDATE Patients

SET city = 'Ernakulam’

WHERE patient_id = 1;


SELECT * FROM Patients;
```

	patient_id	name	age	gender	city
1	1	Rahul Nair	25	Male	Ernakulam
2	2	Anita Sharma	30	Female	Delhi
3	3	Vivek Kumar	45	Male	Chennai
4	4	Sneha Menon	28	Female	Trivandrum
5	5	Arjun Singh	50	Male	Mumbai
6	6	Divya Reddy	35	Female	Hyderabad
7	7	Kiran Patel	40	Male	Ahmedabad
8	8	Meera Das	22	Female	Kolkata
9	9	Suresh Babu	60	Male	Madurai
10	10	Neha Kapoor	27	Female	Chandigarh
11	11	Manoj Pillai	33	Male	Thrissur
12	12	Lakshmi Iyer	48	Female	Coimbatore
13	13	Amit Verma	38	Male	Bangalore
14	14	Priya Joshi	29	Female	Pune
15	15	Rohit Mehta	55	Male	Jaipur

Query 24 :-

- Delete Information

```
DELETE FROM Appointments

WHERE appointment_id = 1008;


SELECT * FROM Appointments;
```

	appointment_id	patient_id	doctor_id	appointment_date	diagnosis
1	1001	1	101	2025-02-01	Chest Pain
2	1002	2	102	2025-02-02	Skin Allergy
3	1003	3	103	2025-02-03	Fracture
4	1004	4	104	2025-02-04	Fever
5	1005	5	101	2025-02-05	Heart Checkup
6	1006	6	105	2025-02-06	Migraine
7	1007	7	103	2025-02-07	Back Pain